XTEND Guidance Engineer Test Submission – Yuval Grimberg

Introduction

This submission addresses both parts of the Guidance Engineer test.
For Q1, I reconstructed the target line-of-sight angles in the world frame (NWU) from the recorded camera measurements, drone attitude, and gimbal pitch, and then estimated the azimuth and elevation rates.
For Q2, I built a 3D interception simulation between a target drone and an interceptor drone under the required speed and acceleration constraints, and evaluated whether interception was achieved.

Q1 – Target Angles in World Frame

Task

The first task was to compute the target azimuth and elevation in the world frame (NWU) from the dataset, using the onboard camera target angles, the drone attitude quaternion, and the gimbal pitch angle. In addition, the task required estimating and plotting the azimuth and elevation rates while balancing noise reduction and low latency.

Implementation

I treated the dataset as an asynchronous sensor log rather than a perfectly synchronized table. The camera measurements were used as the master timeline, since the line-of-sight target angles are defined there. I separated the data into camera, gimbal, and navigation streams, interpolated the gimbal pitch in time, and interpolated the attitude quaternion using SLERP so that all signals could be evaluated consistently at the camera timestamps.

The measured camera azimuth and elevation were converted into a line-of-sight unit vector in the camera frame. That vector was then rotated through the gimbal pitch transformation and through the drone attitude into the world frame, which allowed reconstruction of the target azimuth and elevation in NWU. The implementation also evaluates sign and frame-convention candidates and selects the convention that best matches the data diagnostics. In the current repo notes, the selected convention is: quaternion interpreted as world-to-body, camera azimuth positive right, camera elevation positive down, and gimbal pitch positive down.

To estimate the angular rates, I compared several derivative methods: gradient, Savitzky–Golay, local polynomial, and spline. The reported rate was selected using a weighted score that prioritizes low lag while also considering reconstruction error, smoothness, edge behavior, and holdout performance. According to the repo notes, the selected method for this log was the Savgol (savitky-golay).

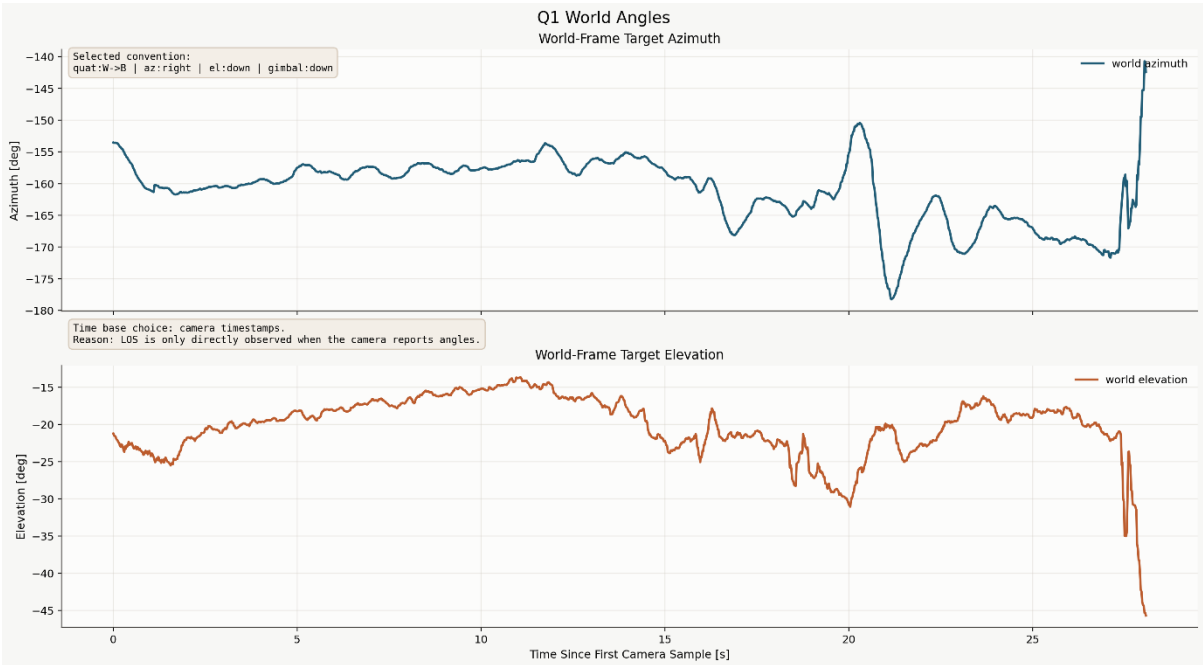
How it works

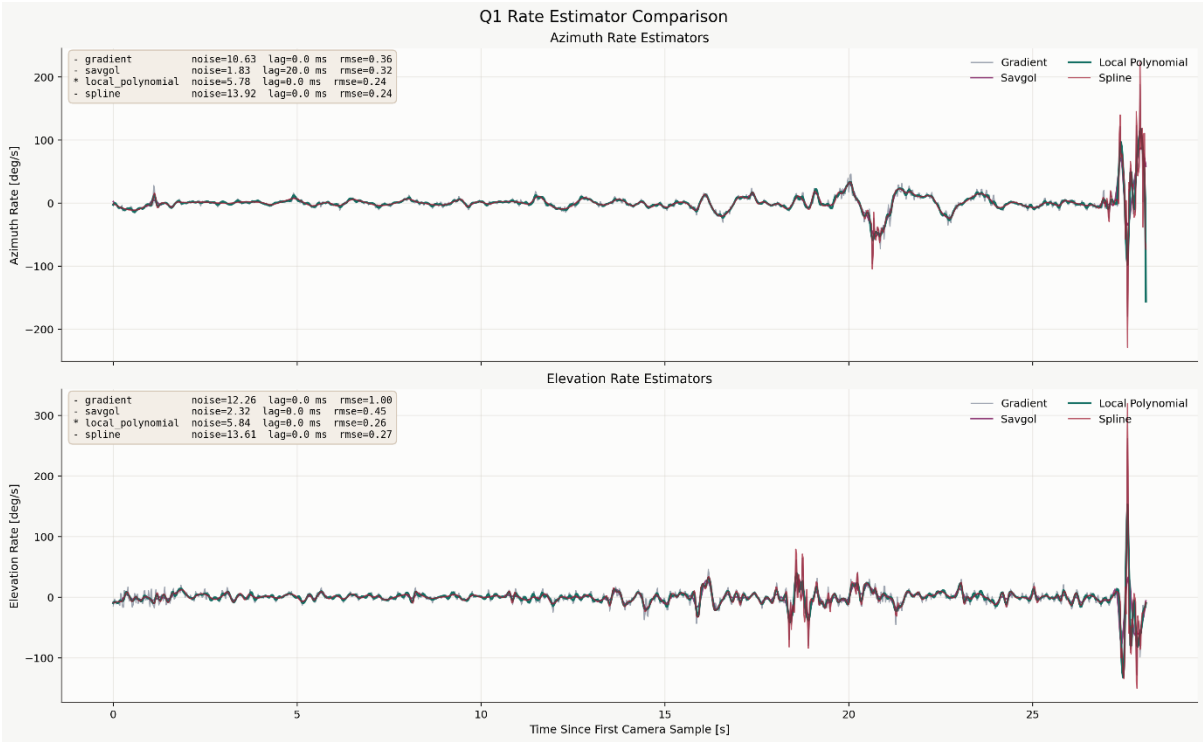
In compact form, the Q1 pipeline is:

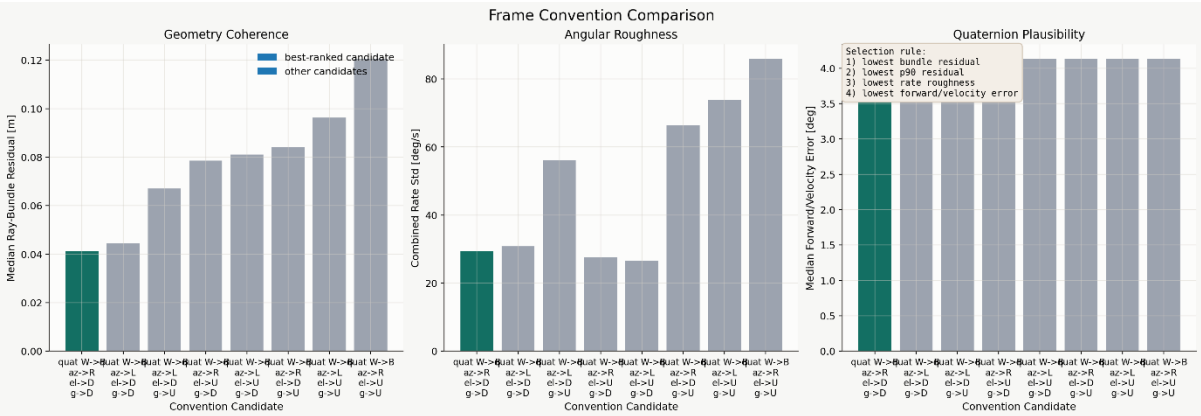
1. Read the dataset and split it into the relevant asynchronous streams.
2. Use the camera timestamps as the reference time base.
3. Interpolate gimbal pitch and attitude to the camera timestamps.
4. Convert camera azimuth/elevation into a 3D LOS vector.

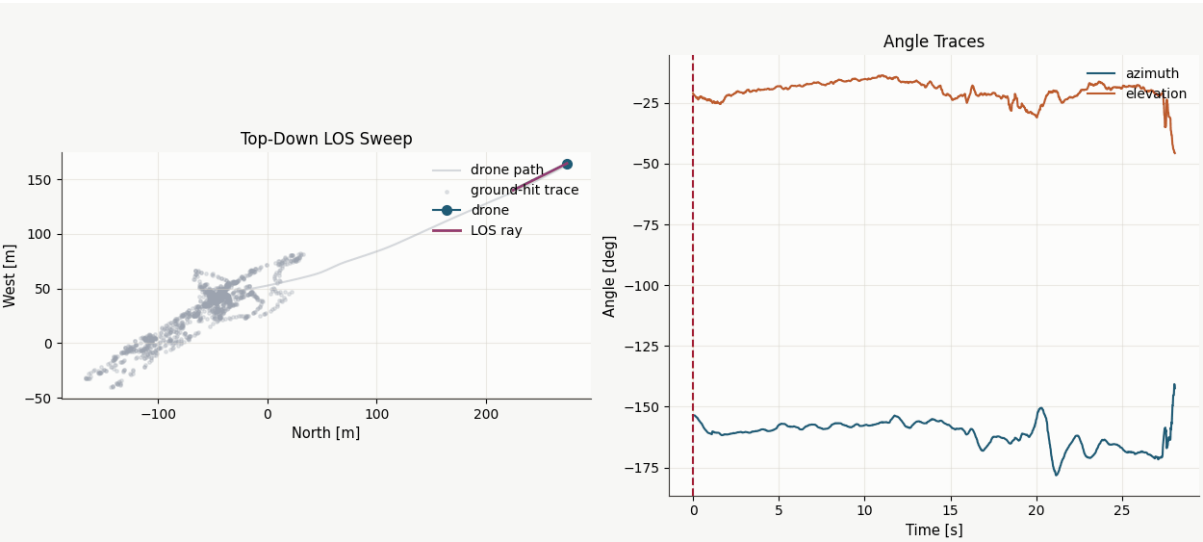
5. Rotate the LOS vector from camera frame to body frame and then to NWU world frame.
6. Extract world-frame azimuth and elevation from the reconstructed LOS vector.
7. Estimate azimuth and elevation rates using candidate filters and select the best one based on the evaluation score.

Plots for Q1









Task

The second task was to simulate a 3D interception scenario in which a target drone starts at $[800,300,200]$ with constant speed13 $\frac{m}{s}$, while an interceptor starts at $[0, 0, 0]$.

The interceptor must obey a maximum speed of 20 $\frac{m}{s}$ maximum acceleration of 2.5 $\frac{m}{s^2}$ along the World X axis and 1.0 $\frac{m}{s^2}$ along the Y axis. The goal is to determine whether interception occurs and, if so, report the intercept time.

Implementation

I built a time-domain 3D simulation of both vehicles in the NWU frame. The target motion was modeled at constant speed, and the interceptor was guided using a predictive interception baseline while enforcing the requested kinematic constraints numerically. In the repo, additional guidance/target-mode comparisons were also generated, but the assignment-aligned baseline is the straight-target predictive case.

he interception logic is based on relative geometry: the interceptor computes a commanded motion toward a future intercept condition rather than only chasing the current target position. At each simulation step, the commanded velocity and the resulting acceleration are clipped so that the speed and the world-frame X/Y acceleration constraints are never violated. The repo notes state that the baseline predictive case achieved interception, with an intercept time of 134.4 s and a minimum separation of about 1.997 m.

Mathmatical models:

A compact way to describe the method is:

$$r(t) = P \cdot T(t) - P \cdot I(t)$$

where $\mathbf{p}_T$ is the target position and $\mathbf{p}_I$ is the interceptor position. The interceptor tries to drive this relative position to zero, or below the chosen intercept radius.

For predictive interception, the commanded direction is based on a future target position estimate rather than the current instantaneous LOS only. In practice, that means the interceptor aims toward a predicted meeting point:

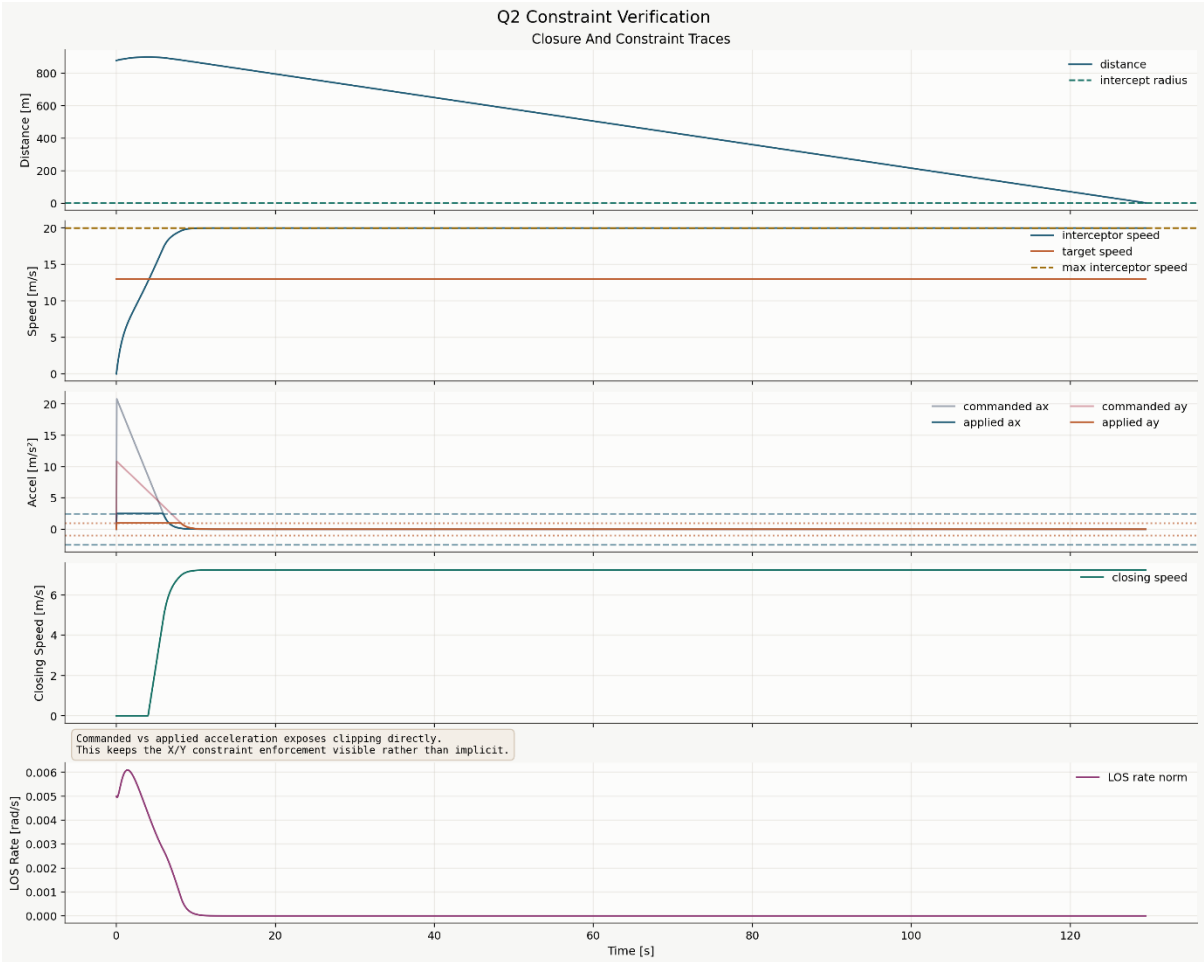
$$pT, pred = pT + vT \cdot tgo$$

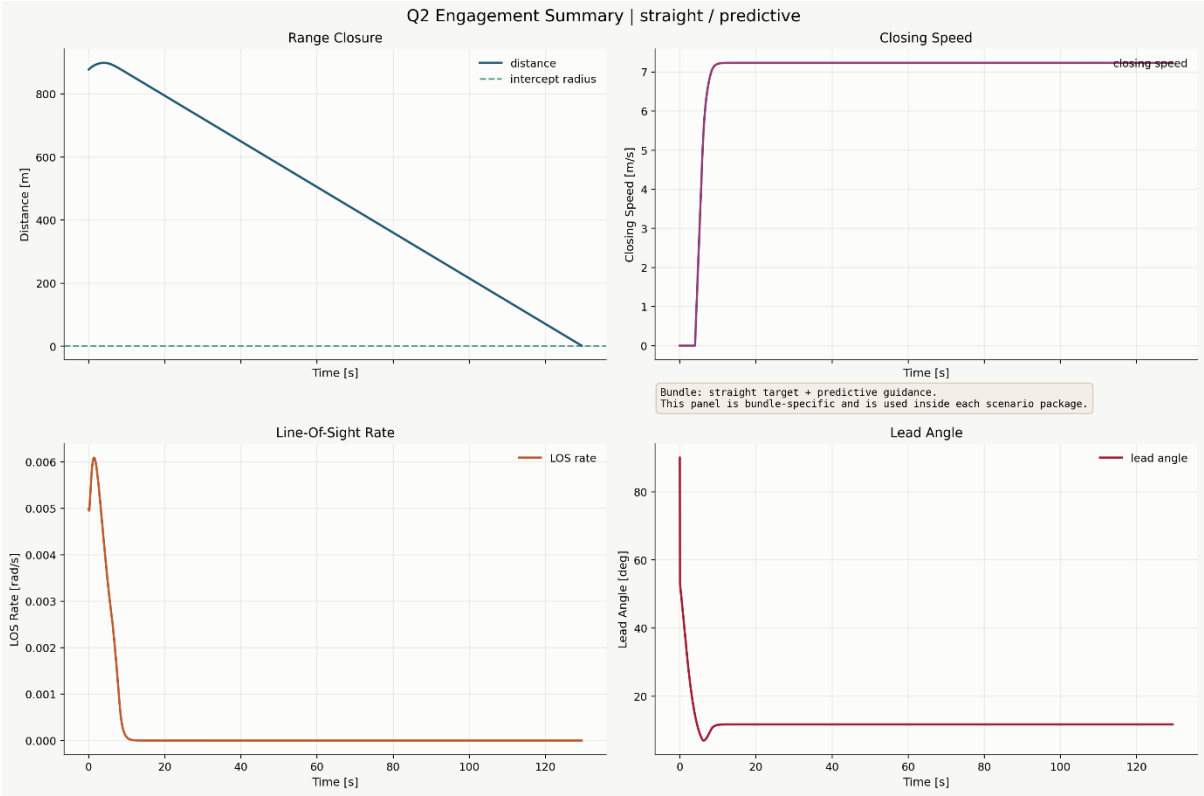
and then generates a velocity command toward that point. The resulting acceleration command is limited component-wise according to the assignment:

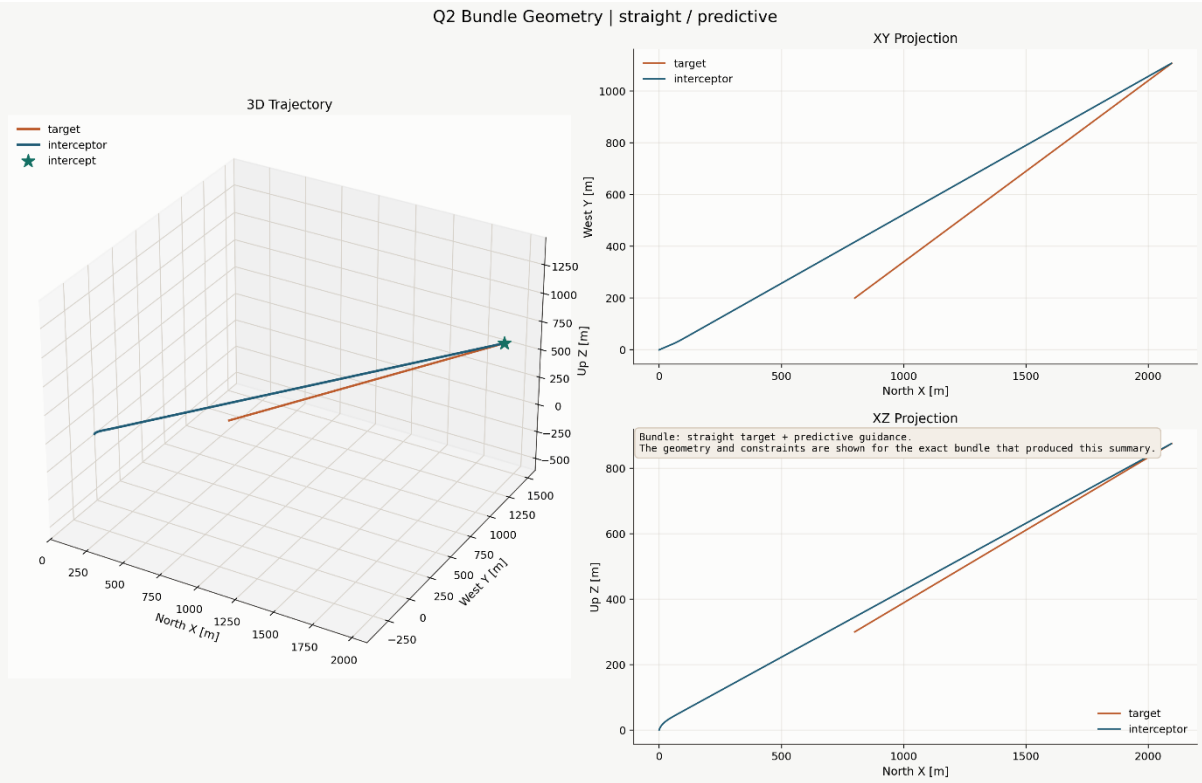
$$|a_x| \leq 2.5 \; and \; |a_y| \leq 1.0 \; also \; ||V_I|| \leq 20$$

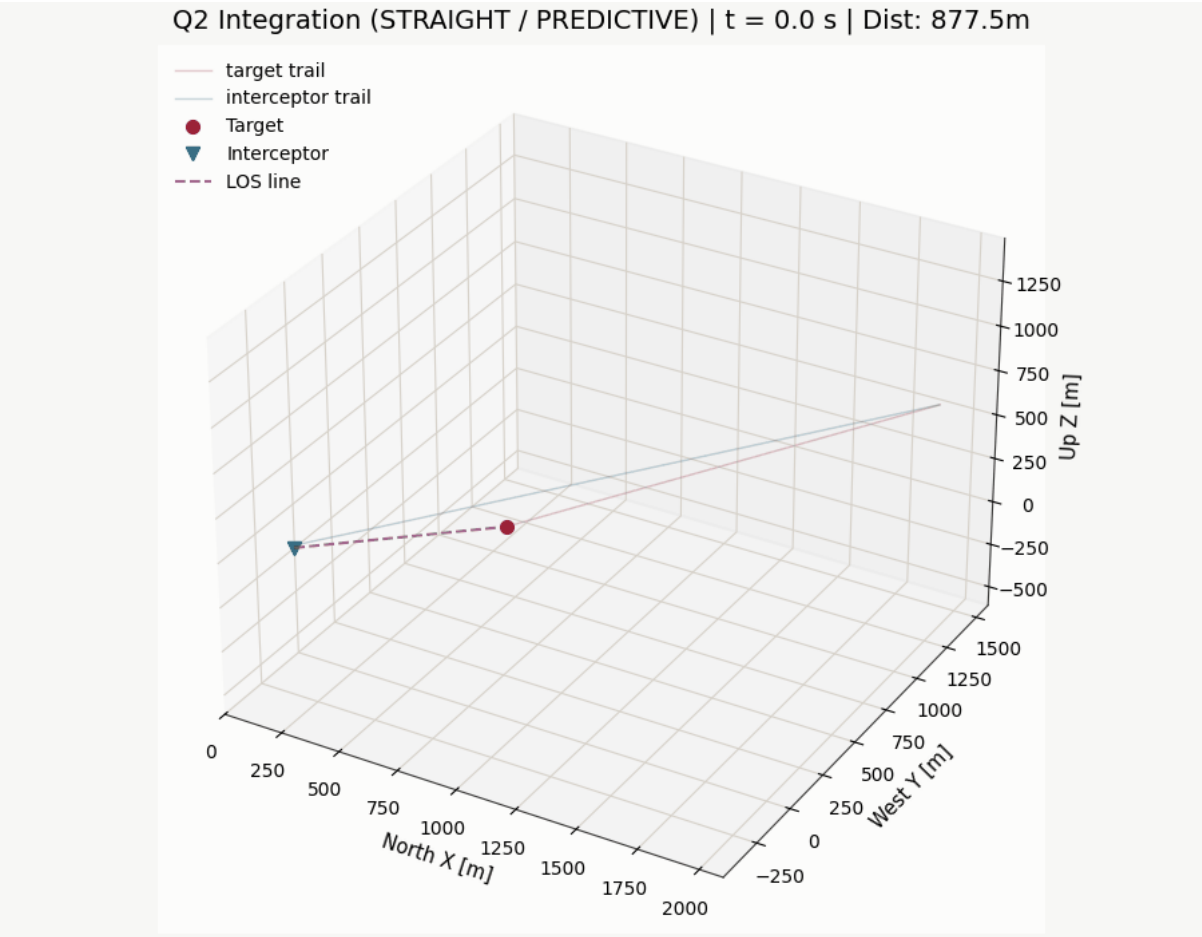
This makes the simulation physically constrained instead of allowing an unrealistically aggressive pursuit law. The repo also notes that no Z-acceleration limit was enforced because the prompt does not specify one.

To sum things up For Q2, I implemented a 3D interception simulation in the NWU frame for both the target and the interceptor. The target starts from the required initial position and moves at a constant speed of 13 m/s, while the interceptor starts from the origin and is constrained by a maximum speed of 20 m/s and bounded accelerations along the world X and Y axes. I used a predictive guidance baseline, meaning that the interceptor aims toward a future intercept condition rather than only the target's current position. At each simulation step, the interceptor command is clipped so that both the speed constraint and the acceleration constraints are satisfied. In the assignment-aligned baseline case, the simulation achieved interception, with a reported intercept time of 134.4 s.


Q2 Constraint Verification
Closure And Constraint Traces
Distance [m]
distance
intercept radius
Speed [m/s]
interceptor speed
target speed
max interceptor speed
Accel [m/s²]
commanded ax
applied ax
commanded ay
applied ay
Closing Speed [m/s]
closing speed
Commanded vs applied acceleration exposes clipping directly.
This keeps the X/Y constraint enforcement visible rather than implicit.
LOS Rate [rad/s]
LOS rate norm
Time [s]


Q2 Engagement Summary | straight / predictive
Range Closure
Distance [m]
distance
intercept radius
Time [s]
Closing Speed
Closing Speed [m/s]
closing speed
Time [s]
Bundle: straight target + predictive guidance.
This panel is bundle-specific and is used inside each scenario package.
Line-Of-Sight Rate
LOS Rate [rad/s]
LOS rate
Time [s]
Lead Angle
Lead Angle [deg]
lead angle
Time [s]





Verdict and Assumptions

Overall, both parts of the assignment were completed with a solution focused on clarity, reproducibility, and physically consistent behavior. For Q1, the main challenge was that the dataset behaves as an asynchronous sensor log, so correct reconstruction required time alignment and careful handling of frame conventions. The final implementation reconstructs the target azimuth and elevation in NWU and estimates the angular rates using a method selected to balance noise reduction and latency. For Q2, the interceptor simulation satisfies the requested speed and acceleration limits while still achieving interception in the assignment baseline case.

The main assumptions were the frame-sign convention selection in Q1 and the absence of direct target range truth in the dataset, which means the geometric LOS diagnostics are used mainly for plausibility and consistency checks rather than absolute validation. In Q2, the prompt does not define a vertical acceleration limit, so only the stated X and Y acceleration limits were enforced directly. In addition, the repository contains some explicit extensions beyond the baseline assignment scenario, but the straight-target predictive case was kept as the main reference result for the submission.

Possible improvements would be to validate the frame convention against hardware documentation, add a more explicit sensor timing model for Q1, and extend Q2 with a more realistic vehicle dynamics model rather than a kinematic velocity-shaping baseline. These would improve physical fidelity, but the current solution is already sufficient for the requested assignment scope.

Conclusion

This submission provides a complete solution for both tasks: reconstruction of world-frame target angles and angular rates from the provided dataset, and a constrained 3D interceptor simulation for the interception problem. The implementation is organized so that the main scripts generate both the numerical outputs and the plots required for review.

Github repo includes more complex solution as well as several guidance laws and harder interception scenarios.

https://github.com/grimgrimberg/XTEND-GNC/tree/master